

Self-Learning: 80386 Memory Management & MESI Cache Protocol — MPCA Module 5

Definition: The 80386 Memory Management Unit (MMU) provides two-stage address translation (**Segmentation & Paging**), while multiprocessor systems maintain cache consistency using the **MESI (Illinois) Protocol**.

1. 80386 Protected Mode Memory Translation

```

Logical Address (Selector:Offset)
    |
    v
[ SEGMENTATION UNIT ] ---> Checks Descriptor in GDT/LDT -> Produces 32-bit Linear Address
    |
    v
[ PAGING UNIT (PG=1) ] ---> Uses CR3, Page Directory & Page Table -> Produces 32-bit Physical Address
  
```

1. Descriptor Tables (GDT, LDT, IDT):

- **GDT (Global Descriptor Table):** Holds segment descriptors accessible by all system tasks.
- **LDT (Local Descriptor Table):** Holds segment descriptors private to an individual application task.
- **IDT (Interrupt Descriptor Table):** Holds Gate Descriptors for 256 interrupt vectors.

2. Two-Level Paging Scheme (4 KB Pages):

A 32-bit Linear Address is split into:

31	22 21	12 11	0
+-----+-----+-----+-----+			
Directory Index	Table Index	Offset in Page	
(10 bits: 1024)	(10 bits: 1024)	(12 bits: 4096 bytes)	
+-----+-----+-----+-----+			

- **CR3** holds base address of **Page Directory**.
- Directory entry points to **Page Table**.
- Page Table entry points to physical 4 KB **Page Frame** in RAM.

2. MESI Cache Coherence Protocol

In symmetric multiprocessing (SMP) systems with private L1/L2 caches, every cache line exists in one of **four MESI states**:

M - MODIFIED : Cache line is dirty (present only in current cache, modified, memory is stale).
E - EXCLUSIVE : Cache line is clean (present only in current cache, matches main memory).
S - SHARED : Cache line is clean (present in multiple processor caches, matches main memory).
I - INVALID : Cache line is invalid (contains obsolete/stale data; must fetch from bus).

State Transition Summary:

- **Read Hit:** Stays in M, E, or S state.
- **Write Hit on S:** Broadcasts invalidate to bus, transitions from **S -> M**.
- **Snooping on Bus Write:** Transitions from **S -> I** or **E -> I**.

3. Quick Recall Flow

Logical (Selector:Offset) -> GDT/LDT -> Linear Address -> Page Dir -> Page Table -> Physical Address |
MESI (Modified, Exclusive, Shared, Invalid)